

Keywords: visual token pruning • multimodal LLMs • conditional removability • inference efficiency

CoRePrune

From Token Importance to Conditional Removability: Rethinking Visual Token Pruning in Multimodal Large Language Models

By Shengli He et al.

arXiv:2609.26484 • September 2026

Token importance and redundancy are poor proxies for removability because they ignore representation depth and deletion-set context; CoRePrune’s progressive depth-aware perturbation tracking and set-conditioned refinement retain 90.3% of dense-model performance with 128 visual tokens while cutting prefill time by 51% across five multimodal LLM backbones.

AT A GLANCE

Problem: Visual token pruning methods score tokens by importance or redundancy but miss that removability is conditioned on transformer depth and the surrounding deletion set.

Why it matters: Incorrect removability estimates cause unexpected quality drops at aggressive budgets; fixing the estimate recovers performance.

Method: Two-stage CoRePrune: progressive perturbation-aware visual pruning across depth, followed by set-conditioned refinement after visual-text interaction.

Result: 90.3% dense-model performance with 128 tokens; 51.0% prefill speedup on five MLLM backbones.

Evidence: Standard image, high-resolution, and video benchmarks across LLaVA-1.5, InternVL2, LLaVA-HR, and VideoLLM.

The Big Picture

Multimodal large language models (MLLMs) convert images into sequences of visual tokens—typically hundreds to thousands—that dominate the cost of the prefill phase. Training-free token pruning addresses this by scoring each visual token and discarding the lowest-scoring ones before or during the forward pass. The appeal is obvious: no fine-tuning, no architecture change, just a cheaper forward pass.

The trouble is that existing scoring functions—attention weight, activation norm, pairwise similarity—treat removability as a static per-token property. CoRePrune shows that it is not. Removing the same token at layer 5 versus layer 20 causes substantially different output perturba-

tions because the representations have been increasingly fused with context. Removing a token when 100 others are already gone has a different effect than removing it from the full set because the remaining tokens have compensated for prior deletions. These two dependencies—*depth* and *deletion set*—are precisely what existing methods ignore.

Why Existing Approaches Fall Short

Importance-based methods (attention score, gradient norm) score tokens in isolation and at a single layer; they cannot predict how importance changes as representations propagate through depth.

Redundancy-based methods (cosine similarity) identify tokens that are similar to their neighbours, but spatial similarity at one layer does not imply similarity after additional self-attention mixing.

Static depth assignment: Most methods apply pruning at a fixed layer (early or late), ignoring that the same token may be safe to remove at depth 20 but harmful to remove at depth 5.

Set-agnostic scoring: All existing methods score each candidate independently, ignoring that removing token v_i with v_j already gone may have a very different effect than removing v_i from the full set.

Core Mechanism

Conditional Removability. CoRePrune operationalises removability as the downstream perturbation caused by removing a token or set of tokens, measured at the current depth under the current deletion set. This quantity is tracked dynamically rather than estimated once at a fixed layer.

Stage 1: Progressive Perturbation-Aware Visual Pruning (PPVP). As the forward pass proceeds through successive transformer layer blocks, PPVP re-estimates the perturbation each candidate removal would cause. A token whose removal perturbation *shrinks* as depth increases has been compensated for by the model’s own representation mixing—it is safe to remove. Tokens whose perturbation grows are promoted in priority.

Stage 2: Set-Conditioned Refinement (SCR). After the first visual-text interaction layer (where visual and text tokens first attend to each other), SCR re-evaluates each remaining candidate under the *current* deletion set. Tokens that now provide net benefit given what has already been removed are recovered; tokens whose marginal contribution has collapsed are added to the deletion set.

Technical Formulation

Let $V = \{v_1, \dots, v_N\}$ be the visual token sequence and $f_l(S)$ the representations at layer l for token set S .

Perturbation at depth l for deletion set D :

$$P_l(D) = \|f_l(V \setminus D) - f_l(V)\|_F$$

PPVP priority update for candidate v_i :

$$\text{safe at } l_2 \iff P_{l_2}(\{v_i\}) < P_{l_1}(\{v_i\}), \quad l_2 > l_1$$

SCR marginal contribution under current deletion set D at layer l^* :

$$r(v_j; D) = P_{l^*}(D) - P_{l^*}(D \cup \{v_j\})$$

Recover v_j if $r(v_j; D) > 0$; add v_j to D if $r(v_j; D) \ll 0$.

Learning or Inference Procedure

CoRePrune is **training-free** and runs during the prefill phase:

1. Forward pass to layer l_1 : compute initial perturbation estimates.
2. Continue to layer $l_2 > l_1$: update estimates via PPVP; rank candidates.
3. Tentatively remove lowest-priority tokens to meet the token budget.
4. Continue to visual-text interaction layer l^* .
5. Apply SCR: recover tokens with $r(v_j; D) > 0$; finalise deletion set.
6. Continue forward pass with the pruned set; generate output.

Overhead: two additional perturbation probes during prefill, negligible relative to the attention savings from the reduced token count.

What the Guarantee Says

PPVP provides a constructive argument: if $P_{l_2}(\{v_i\}) < P_{l_1}(\{v_i\})$, the model has already compensated for v_i 's absence by depth l_2 , making removal safe. SCR provides a greedy local-optimum guarantee: after SCR, no single token swap (recovery or addition) would reduce the deletion-set perturbation.

Experimental Findings

- **5 MLLM backbones:** LLaVA-1.5, InternVL2, LLaVA-HR, VideoLLM, and one additional model
- At **128 visual tokens:** CoRePrune retains **90.3%** of dense-model performance (vs. 83–86% for prior methods) and reduces aggregate prefill time by **51.0%** (vs. 45–48% prior)

- At 64 tokens: advantage over baselines grows due to stronger deletion-set interaction effects
- Largest gains on global reasoning tasks (VQA, video QA); smaller gains on local patch tasks where interaction effects are weaker

Ablations and Interpretation

PPVP alone removes the depth-static assumption and recovers half the total gain over attention-based baselines.

SCR alone removes the set-agnostic assumption and recovers the other half. The joint gain exceeds the sum, revealing non-linear interaction.

PPVP probe depth: Tracking across ≥ 3 layer blocks is necessary; fewer blocks converges to depth-static behaviour.

SCR timing: Applying SCR before visual-text interaction or at the final layer both degrade performance; the visual-text fusion layer is the optimal intervention point.

Reference

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<https://arxiv.org/abs/2609.26484>